Geo/A2/Trig Finals

Semester 1

Overview & Plan

- Review class notes first
  - Do a chapter by chapter breakdown based on which concept you don't understand the most
- For each chapter go over key theorems and definitions
  - Take mini practice tests to help yourself study in a timed environment
- Spend time on your own going over each section's theorems and understanding each theorem's givens and conclusion.

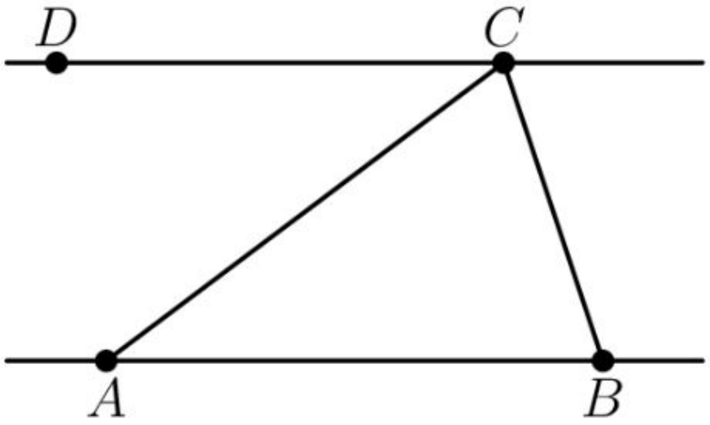
Applying what you know: Euler Line

The Euler line of a triangle includes such “special points” as the orthocenter, circumcenter, and centroid. For the triangle with A(a1, a2), B(b1, b2), C(c1, c2),

a. Use two “special points” to find the equation
b. Verify that the third “special point” lies on the Euler line as well.

If this is too hard, just pick random points that form a (non-equilateral triangle)

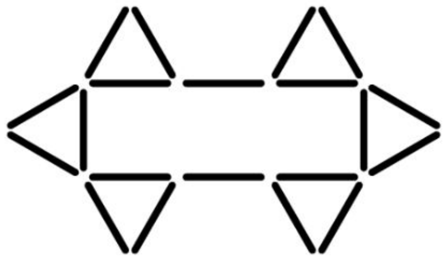
In the figure, $\overline{DC}$ is parallel to $\overline{AB}$. We have $\angle DCA = 40^\circ$ and $\angle ABC = 73^\circ$. Find $\angle ACB$.



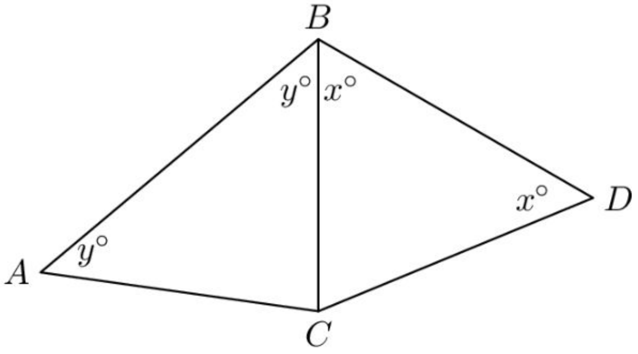
The measures of angles A and B are both positive, integer numbers of degrees. The measure of angle A is a multiple of the measure of angle B, and angles A and B are complementary angles. How many measures are possible for angle A?

___

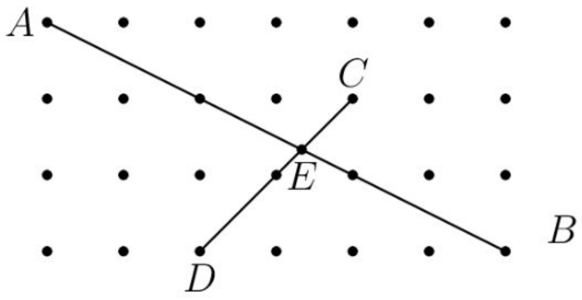
Twenty toothpicks, each 6 cm long, form a figure consisting of six equilateral triangles and a rectangle as shown. What is the total area of the figure? Express your answer as a decimal to the nearest hundredth.



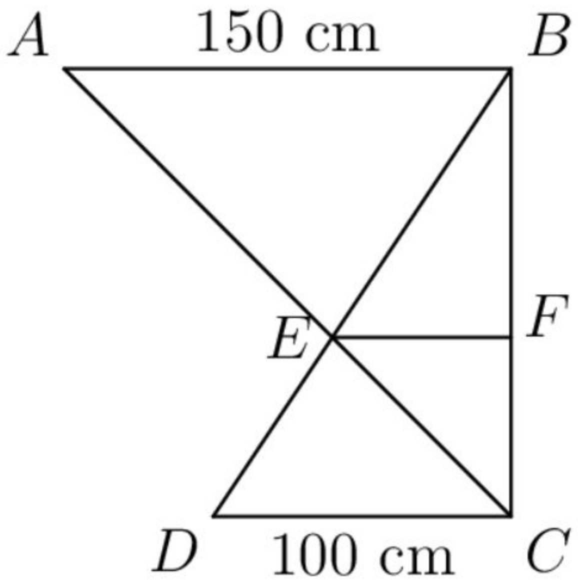
In the diagram, triangles ABC and CBD are isosceles. The perimeter of $\triangle CBD$ is $19,$ the perimeter of $\triangle ABC$ is $20,$ and the length of BD is $7.$ What is the length of $AB?$



The diagram shows 28 lattice points, each one unit from its nearest neighbors. Segment AB meets segment CD at E. Find the length of segment AE.



What is the number of centimeters in the length of EF if $AB \parallel CD \parallel EF$?

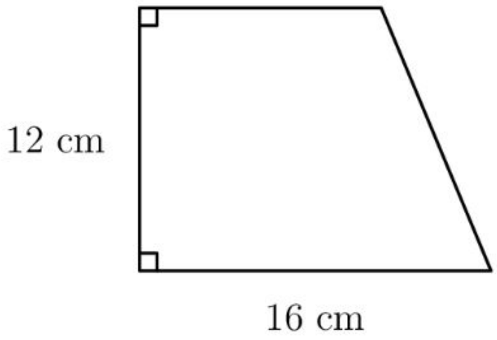


The side lengths of a right triangle measure x, $x+y$ and $x-y$ units where $0 < y < x$. What is the value of $y \div x$? Express your answer as a common fraction.

What is the radius of the circle inscribed in triangle ABC if $AB = 12,\ AC = 14,\ BC = 16$?

In triangle ABC, $AB = 5$, $BC = 8$, and the length of median AM is 4. Find AC.

The trapezoid shown has a height of length 12 cm, a base of length 16 cm, and an area of 162 cm^2. What is the perimeter of the trapezoid?

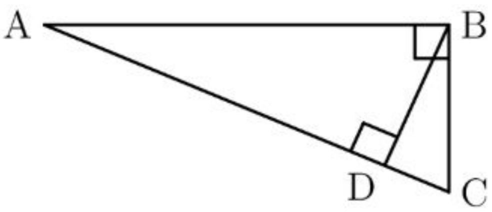


Define $a \clubsuit b = a^2b - ab^2$. Which of the following describes the set of points (x, y) for which $x \clubsuit y = y \clubsuit x$? Enter the capital letter corresponding to the correct answer choice.

(A) a finite set of points
(B) one line
(C) two parallel lines
(D) two intersecting lines
(E) three lines

What is the radius of the circle inscribed in triangle ABC if $AB = 12,\ AC = 14,\ BC = 16$? Express your answer in simplest radical form.

In the figure shown, $AC = 13$ and $DC = 2$ units. What is the length of the segment BD? Express your answer in simplest radical form.



A rhombus has sides of length 51 units each and a shorter diagonal of length 48 units. What is the length, in units, of the longer diagonal?

___

Four diagonals of a regular octagon with side length 2 intersect as shown. Find the area of the shaded region.

Slide deck QR Code: